

IGOR Final Report

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Executive Summary

This research initiative aimed to optimize an in-house PURE cell-free protein synthesis system by simultaneously maximizing protein yield (sigmoid_steady_state) and reaction rate (sigmoid_rate). The IGOR process systematically analyzed existing literature and experimental data to formulate and refine hypotheses testable within a Bayesian optimization framework. The key outcome was the development of a powerful conceptual model, "Pareto Frontier of Metabolic Efficiency," which frames the optimization problem as a predictable trade-off between two metabolic regimes: a high-rate, low-yield 'sprint' and a low-rate, high-yield 'marathon'. This trade-off is hypothesized to be controlled by the concentration ratio of energy sources (ATP, GTP) to biosynthetic substrates (amino acids, tRNA). This framework provides a clear, actionable strategy for the next round of multi-objective, AI-guided experiments to map the system's performance limits.

Top-Ranked Hypotheses

1: Pareto Frontier of Metabolic Efficiency: Defining the 'Sprint vs. Marathon' Trade-Off in PURE Systems

Hypothesis: Hypothesis: The optimal performance of the PURE system lies on a Pareto frontier defined by the competing objectives of reaction rate (sigmoid_rate) and final yield (sigmoid_steady_state). The position along this frontier is controlled by the ratio of high-energy phosphates ([ATP]+[GTP]) to biosynthetic building blocks ([Amino Acids]+[tRNA]).

Rationale: This hypothesis evolves the winning 'sprint vs. marathon' analogy by formalizing it with the concept of a Pareto frontier. It combines the intuitive framing from 'Explorer 1' with the direct, mechanistic ratio from 'Architect 1'. High energy-to-substrate ratios fuel a 'sprint' by saturating the translational machinery for maximum speed, but lead to rapid accumulation of inhibitory byproducts (e.g., inorganic phosphate) and premature resource depletion, resulting in a

high rate but low yield. Conversely, low, balanced ratios enable a 'marathon' of sustained, efficient synthesis that maximizes total protein output at the cost of initial speed.

Testable Question: Using multi-objective Bayesian optimization, can we map a Pareto-optimal front for rate vs. yield? **Does the $\frac{[ATP]+[GTP]}{[Amino\ Acids]+[tRNA]}$ ratio act as a primary axis for navigating this front, allowing us to predictably select for either a 'sprint' (rate-maximized) or 'marathon' (yield-maximized) metabolic regime?** **Final Rationale:** This hypothesis was ranked highest because it elegantly synthesizes the most effective elements from the preceding hypotheses. It incorporates the directly testable, mechanistic core of its competitors by focusing on the energy-to-substrate ratio, a variable directly aligned with the experimental plan. Crucially, it retains the 'sprint vs. marathon' analogy, which was consistently favored in debates for providing a powerful and intuitive conceptual framework for a multi-objective optimization problem. This formalization into a Pareto frontier makes the hypothesis both scientifically rigorous and highly effective for communicating and guiding the research. **Key Supporting Evidence:** - Rationale from 'Explorer 1' vs 'Architect 1' debate: The 'sprint vs. marathon' analogy provides a powerful and intuitive framework for interpreting multi-objective optimization results. - Finding: PURE system productivity is often limited by the depletion of energy and nutrient molecules, such as ATP, GTP, and amino acids, and the accumulation of inhibitory byproducts like inorganic phosphate. (Li et al., 2014; Y. Shimizu et al., 2001) - Opportunity: There is a gap in understanding the explicit trade-offs between maximizing reaction rate ('sigmoid_rate') and maximizing final yield ('sigmoid_steady_state') through the simultaneous tuning of energy and substrate concentrations.

2: Resource Allocation Theory for Cell-Free Synthesis: A Trade-off Between Catalytic Speed and Substrate Efficiency

Hypothesis: Hypothesis: The PURE system's performance is governed by a fundamental trade-off in the allocation of finite catalytic resources (e.g., ribosomes, enzymes) towards two competing metabolic strategies: maximizing the speed of synthesis versus maximizing the overall efficiency and final yield from available substrates. The relative concentrations of energy carriers (ATP, GTP) versus building blocks (amino acids, tRNA) dictate this allocation.

Rationale: This hypothesis evolves the core mechanism of 'Architect 1' by applying an out-of-box concept from Resource Allocation Theory in ecology. High concentrations of ATP/GTP relative to substrates signal an environment of abundant energy, promoting a 'fast growth' (high rate) strategy that is rapid but potentially wasteful. Conversely, when

substrates are more abundant relative to the energy supply, the system adopts a 'resource conservation' (high efficiency) strategy, slowing down the rate to maximize the total protein yield (sigmoid_steady_state) from the available building blocks.

Testable Question: Does varying the ratio of energy sources to biosynthetic substrates cause a predictable shift in resource allocation, evidenced by distinct, non-overlapping optima for reaction rate and final yield? Can we model this behavior to identify a 'tipping point' in the resource ratio that causes the system to switch from a rate-optimized to a yield-optimized state? **Final Rationale:** This hypothesis was a very strong contender due to its solid mechanistic foundation and direct testability within the defined experimental space. Its application of Resource Allocation Theory provides a novel and compelling lens for interpreting system behavior. While it was ultimately edged out by the 'Pareto Frontier' hypothesis due to the latter's slightly more intuitive and communicative framing, this hypothesis remains highly valuable as it provides a complementary, theory-grounded perspective on the same core scientific question. **Key Supporting Evidence:** - Rationale from 'Architect 1' vs 'Quant 1' debate: Winning hypotheses must be directly and immediately testable within the defined experimental design space. - Finding: A rapid translation rate can deplete the available energy molecules faster than the regeneration system can keep up, negatively impacting overall yield. - Finding: The concentrations of ATP and GTP are critical for energizing multiple steps of translation, and their regeneration is a key factor in sustaining protein synthesis over time.

Proposed Experimental Plan

Objective

To empirically test the 'Pareto Frontier of Metabolic Efficiency' hypothesis by mapping the trade-off between reaction rate (sigmoid_rate) and final yield (sigmoid_steady_state) in the in-house PURE system. The primary goal is to determine if the ratio of energy sources to biosynthetic substrates controls the system's position on this frontier.

Methodology

A multi-objective Bayesian optimization approach will be employed to efficiently explore the high-dimensional parameter space.

Experimental Setup

1. System Configuration:

- **Reporter:** deGFP (isplamGFP = 0)
 - **PURE System:** In-house mix (IsNEB = 0)
 - **Reagents:** Use confirmed-quality magnesium acetate stock (HasMgAR953 = 0)
2. **Variable Parameters & Bounds:**
- **Primary Variables (Hypothesis-Testing) :**
 - [ATP]: 1 mM - 4 mM
 - [GTP]: 1 mM - 4 mM
 - [Amino acid mix]: 0.1 mM - 0.6 mM
 - [tRNA]: 1 ug/uL - 5.25 ug/uL
 - **Secondary Variables:** All other non-constant parameters from 20260309-round-1-bounds.csv will be co-varied by the optimization engine within their specified min and max bounds.
3. **Experimental Design & Execution:**
- Configure the Bayesian Optimization engine with the multi-objective goal of simultaneously maximizing sigmoid_steady_state and sigmoid_rate.
 - Execute the 10 Generated Experimental Suggestions (GESs) provided by the FWmai platform. These conditions are computationally designed to optimally explore the parameter space and identify points along the putative Pareto frontier.
 - **Control Reaction:** Include a baseline control experiment with standard component concentrations (e.g., ATP=2mM, GTP=2mM, Amino Acids=0.35mM, tRNA=3.5ug/uL) to serve as a benchmark for performance comparison.

Data Collection & Analysis

1. **Measurement:** For each experimental condition, measure deGFP fluorescence over time using a plate reader.
2. **Data Processing:** Fit the kinetic fluorescence data for each successful reaction to a sigmoidal curve to derive the key metrics: sigmoid_steady_state (ng/uL) and sigmoid_rate (1/h).
3. **Hypothesis Validation:**
 - Generate a 2D scatter plot of sigmoid_rate vs. sigmoid_steady_state for all experimental results to visualize the performance landscape.
 - Identify the set of non-dominated points that form the experimental Pareto frontier.
 - For each experimental point, calculate the energy-to-substrate ratio: $([ATP] + [GTP]) / ([Amino\ acid\ mix] + [tRNA])$ (ensure consistent units for ratio calculation).
 - Color-code the points on the scatter plot according to this ratio to determine if it correlates with the position along the Pareto front.

Expected Outcome

The results are expected to form a distinct curve representing the Pareto-optimal front, demonstrating a clear trade-off between rate and yield. Conditions with a high energy-to-substrate ratio are predicted to populate the high-rate, lower-yield region ('sprint' regime), while conditions with a low, balanced ratio will populate the lower-rate, high-yield region ('marathon' regime), thus validating the hypothesis.

References

- [Improved Cell-Free RNA and Protein Synthesis System, Li, J., 2014, PLoS ONE]
- [Cell-free translation reconstituted with purified components, Shimizu, Y., 2001, Nature Biotechnology]
- [Dissecting limiting factors of the Protein synthesis Using Recombinant Elements (PURE) system, Li, J., 2017, Integrative Biology]
- [Continued Protein Synthesis at Low [ATP] and [GTP] Enables Cell Adaptation during Energy Limitation, Jewett, M.C., 2009, Journal of Bacteriology]
- [Modeling Cell-Free Protein Synthesis Systems—Approaches and Applications, Borkowski, O., 2020, Frontiers in Bioengineering and Biotechnology]

Dead Ends

- Synergistic Dual-Engine Energy Regeneration for Maximizing PURE System Yield: This hypothesis was rejected because its core testable parameters, Polyphosphate ([PolyP]) and Polyphosphate Kinase ([PPK]), were fixed at zero in the defined experimental bounds for the upcoming optimization round. This made the hypothesis untestable and not actionable, despite being well-supported by historical data.